Dimensionality Reduction as Alpha? Comparing Linear and Deep Feature Extractors for Tree-Based Cross-Sectional Return Prediction

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Abstract

The latent-factor hypothesis implies that a high-dimensional library of alpha signals is largely redundant, suggesting that compressing it prior to prediction should improve the signal-to-noise ratio of the inputs without discarding economic content. This thesis examines that proposition. Five upstream feature extractors — the raw alpha library, PCA, Factor Analysis, the GKX Autoencoder, and FactorVAE are evaluated as interchangeable inputs to gradient-boosted ranking models, under a strict, leakage-controlled walk-forward protocol across two asset universes and carried through to the Sharpe ratio of a long-short portfolio. The evidence indicates that dimensionality reduction does not yield a method that improves the performance of the ranking models by a statistically significant margin in both universes. FactorVAE is the notable exception: it is the least unprofitable configuration in both markets and improves upon the baseline in each, though this improvement is statistically significant only on the S&P 500 and not on the ETF universe, where the advantage is economically present but not significant. The realised losses are shown to be market-neutral, loading on the regression intercept rather than on residual market exposure, and are traced to transaction-cost drag: measured turnover averages 1.46 times the gross book per day, and gross of costs the configurations are approximately flat rather than loss-making. FactorVAE's advantage is correspondingly shown to be one of ranking persistence rather than ranking accuracy. The findings suggest that feature compression should be treated as an empirical hypothesis, validated against the economic objective, rather than adopted as a default preprocessing step.

Keywords:dimensionality reduction; cross-sectional return prediction; gradient boosting; autoencoders; walk-forward validation.

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Chapter 1

Introduction

1.1 Motivation

The past decade has witnessed a fundamental transformation in empirical asset pricing. What was once the exclusive domain of linear factor models the Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965), the three-factor model of Fama & French (1993), and their five-factor extension (Fama & French, 2015) has been reshaped by the application of machine learning to the problem of cross-sectional return prediction. The landmark study by Gu, Kelly & Xiu (2020) demonstrated that nonlinear machine learning models, evaluated on a large set of firm characteristics, generate economically significant return forecasts that outperform both linear models and the historical mean a finding that has since accelerated the adoption of data-driven methods across quantitative finance.

Within this shift, two questions have emerged as practically important but empirically underexplored. The first is whether the raw input features presented to a predictive model can be improved through dimensionality reduction before training begins. Alpha signals — quantitative indicators of expected return derived from price, volume, and fundamental data are individually noisy, mutually correlated, and often redundant. There is strong theoretical motivation for this compression. The latent factor hypothesis the proposition that asset returns follow the structure

$$r_{i,t} = \beta_i^\top f_t + \epsilon_{i,t} \tag{1.1}$$

where $r_{i,t}$ is the return of asset i at time t, f_t is a vector of K latent factors with $K \ll N$ (the number of assets), β_i is the corresponding vector of factor loadings, and $\epsilon_{i,t}$ is idiosyncratic noise uncorrelated across assets (Ross, 1976; Chamberlain & Rothschild, 1983) implies that the true dimensionality of the cross-section of returns is low. A large library of alpha signals, each of which partially captures these common drivers, therefore contains substantial redundancy. A low-dimensional representation that discards this redundancy should preserve the predictive signal while suppressing noise, a conjecture supported empirically by Gu, Kelly & Xiu (2021), who show that autoencoder-based compression of stock characteristics outperforms linear factor models in out-of-sample asset pricing.

The second question concerns which class of dimensionality reduction is most appropriate for this task. Linear methods such as Principal Component Analysis (PCA) and Factor Analysis (FA) extract orthogonal linear combinations of the input features and have been standard tools in finance for decades. Deep methods specifically the GKX Autoencoder of Gu, Kelly & Xiu (2021) and the FactorVAE of Duan et al. (2022) offer a more flexible nonlinear alternative, learning latent representations that are not constrained to linear factor structure. Crucially, this thesis evaluates all four reduction methods as upstream *feature extractors* feeding a fixed downstream predictor, rather than as the end-to-end asset-pricing models for which the deep variants were originally proposed. The implicit question across both paradigms is the same: does the added complexity of a deep architecture translate into better-quality features, and ultimately into better out-of-sample portfolio performance?

This thesis investigates these questions systematically. We construct an end-to-end machine learning pipeline built on a curated library of 80 alpha signals derived from the 101 Formulaic Alphas (Kakushadze, 2016), implemented following the open-source notebook of Jansen (2020), and applied to two distinct asset universes using daily price and volume data sourced from Yahoo Finance via the `yfinance` Python library. The first universe consists of 60 Exchange-Traded Funds (ETFs) spanning equity sectors, geographic regions, fixed income, and commodities, with daily data from 2006 to 2025 and a walk-forward out-of-sample evaluation over 2019–2025. The second universe consists of S&P 500 constituents, with data from 2020 to 2025 and an out-of-sample evaluation over 2022–2025.

All configurations are evaluated on a strict walk-forward out-of-sample (OOS) basis. We utilise the Rank Information Coefficient (Rank IC) as a diagnostic of out-of-sample predictive skill, while final model viability

is determined by the Sharpe ratio of the simulated long–short portfolio, which serves as the ultimate economic criterion.

The motivation for choosing gradient boosting as the downstream predictive model is empirical, practical, and data-driven. Tree-based gradient boosting methods have consistently demonstrated state-of-the-art performance on tabular financial data (Gu, Kelly & Xiu, 2020; Chen & Guestrin, 2016), and a growing body of evidence shows that gradient-boosted trees remain better on tabular data, matching or outperforming deep learning architectures on datasets of the size typical in this setting, owing to inductive biases better suited to tabular features (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022). This is a material constraint in our setting: the ETF universe comprises 60 instruments, yielding cross-sectional training windows of the order of tens of thousands of observations per fold a regime in which deep end-to-end predictors are at a structural disadvantage relative to gradient boosting. Applying a deep learning model directly as the forecaster would therefore confound the comparison: any underperformance could reflect insufficient data for the network rather than a genuine inferiority of the learned features.

Beyond the data-size argument, recent work in quantitative finance has established that ranking objectives outperform regression for cross-sectional return prediction. Pairwise and listwise learning-to-rank formulations directly optimise the relative ordering of assets the quantity that determines portfolio construction rather than minimising squared prediction error, which treats all mis-rankings equally regardless of their portfolio impact. Poh et al. (2021) demonstrate that learning-to-rank approaches applied to cross-sectional momentum strategies boost Sharpe ratios approximately threefold over traditional regression-based approaches, with the weakest ranking model still significantly outperforming the best regression benchmark. This finding motivates the inclusion of ranking objectives (`lambdarank` for LightGBM, `YetiRank` for CatBoost).

Finally, gradient boosting imposes no architectural constraint on the upstream feature extractor, making it an ideal, model-agnostic evaluation environment for comparing the dimensionality reduction methods that are the primary subject of this thesis.

1.2 Research Questions

The thesis is organised around three research questions:

Q1. Does compressing the alpha-signal library through dimensionality reduction improve the out-of-sample performance of a tree-based predictor relative to using the raw, uncompressed signals?

Q2. Do deep extractors (the GKX Autoencoder and FactorVAE) produce better-quality features than classical linear methods (PCA and Factor Analysis), and does architectural complexity justify its additional cost?

Q3. How does the choice of feature normalisation — time-series (standard) versus cross-sectional — interact with the reduction method and the breadth of the asset universe?

1.3 Gap in the Literature

Despite the growing body of work on both ML-based return prediction and deep generative feature extraction, a critical gap exists at their intersection: no study has systematically evaluated linear and deep dimensionality reduction methods as upstream feature engineering stages for gradient boosting models in a controlled, walk-forward out-of-sample setting.

First, prior work treats deep DR methods as standalone end-to-end predictors, not as feature extractors. The GKX Autoencoder (Gu, Kelly & Xiu, 2021) was proposed and evaluated as a self-contained conditional asset pricing model that maps stock characteristics directly to latent factors and return predictions. Similarly, FactorVAE (Duan et al., 2022) integrates a variational autoencoder with a factor model in a single architecture that simultaneously extracts latent factors and predicts cross-sectional returns. Neither framework was designed or evaluated as a feature engineering module feeding a separate downstream predictor. The consequence is that it remains unknown whether the latent representations these models produce are useful as general-purpose features, independently of the specific prediction head attached to them.

Second, no existing study performs a controlled, head-to-head comparison of PCA, Factor Analysis, the GKX Autoencoder, and FactorVAE on the same dataset, with the same downstream model and the same evaluation protocol. Existing comparisons are partial: Gu, Kelly & Xiu (2020) compare ML predictors but hold the feature set fixed; Gu, Kelly & Xiu (2021) compare the autoencoder against linear factor models but within the asset pricing framework, not as a feature engineering comparison for a fixed predictor. The survey by Htun,Biehl & Petkov(2023) identifies PCA and autoencoders as among the most effective feature extraction techniques for stock market prediction, but does not conduct a controlled empirical evaluation across a unified pipeline.

Third, the evaluation methodology in much of the existing literature is insufficiently rigorous for financial data. Many studies report in-sample performance, use a single train-test split, or fail to refit the dimensionality reduction within each fold of cross-validation a critical omission that introduces information leakage from the test period into the feature space. Walk-forward validation with strict train-test separation at every preprocessing step is a methodological requirement for credible out-of-sample evaluation in finance, and it has rarely been applied in studies comparing dimensionality reduction methods for alpha signal processing.

This thesis addresses all three dimensions simultaneously: it evaluates DR methods as feature extractors for a fixed downstream predictor, performs a controlled five-way comparison on two asset universes, and enforces strict walk-forward out-of-sample evaluation with preprocessing applied inside each fold to prevent leakage.

1.4 Contributions

This thesis makes the following contributions:

1. **A systematic, controlled comparison of five dimensionality reduction configurations as feature engineering for gradient boosting models.**
   We evaluate Baseline (no reduction), PCA, Factor Analysis, GKX Autoencoder (Gu, Kelly & Xiu, 2021), and FactorVAE (Duan et al., 2022) as upstream feature extractors for LightGBM and CatBoost. To our knowledge, this is the first study to compare all five configurations under a unified, controlled pipeline on financial alpha signals.

2. **A rigorous walk-forward evaluation framework with strict leakage prevention.**
   All preprocessing transformations standardization, PCA, FA, and neural network training for AE and VAE are fitted exclusively on training data within each walk-forward anchor period and applied to the test period without refitting. This eliminates the information leakage that affects many existing studies and ensures that reported performance metrics are genuine out-of-sample estimates.

3. **An evaluation across two asset universes and multiple market regimes.**
   By applying the pipeline to both an ETF universe (out-of-sample 2019–2025) and an S&P 500 universe (out-of-sample 2022–2025), we assess whether findings generalise across asset classes and time periods, including the 2020 COVID-19 crash and the post-COVID regime shift that constitutes one of the most structurally distinct market environments in recent history.

4. **An end-to-end bridge from predictive IC to portfolio economics.**
   All configurations are passed through a portfolio backtester implementing long-short strategies with transaction cost adjustments. This allows us to answer the practically important question of whether statistically significant IC improvements at the prediction stage translate into economically meaningful differences in risk-adjusted portfolio performance.

5. **An analysis of feature normalisation as a first-order design choice.**
   We compare time-series (standard) and cross-sectional normalisation across all configurations and show that the appropriate choice depends jointly on the breadth of the universe and on whether the signal is consumed as a ranking or as an expected-return level.

6. **A robustness analysis establishing market orthogonality and statistical significance.**
   Beyond point estimates, we regress realised long–short returns on the market to confirm that performance is market-neutral, and we test differences in Sharpe ratios and Rank IC against the baseline using block-bootstrap and Newey–West HAC inference.

1.5 Summary of Findings

The central result is that dimensionality reduction is not a free lunch in this setting. Under realistic transaction costs and strict walk-forward evaluation, no reduction method improves on the raw-alpha baseline by a statistically significant margin in both universes, and every configuration produces a negative out-of-sample Sharpe ratio. Architectural sophistication is no guarantee of better features: the deterministic GKX Autoencoder is consistently among the weakest configurations, whereas the regularised FactorVAE, paired with LightGBM, is the least unprofitable in both universes improving upon the baseline by a statistically significant margin on the S&P 500 (block-bootstrap $p < 0.01$), though not significantly on the narrower ETF universe. We further show that the realised losses are market-neutral, loading on the regression intercept rather than on market exposure, and that predictive accuracy is a poor proxy for economic value, since the configuration with the highest Rank IC is frequently not the one with the best Sharpe ratio. The losses are further traced to transaction-cost drag:

measured turnover averages 1.46 times the gross book per day, and gross of costs the configurations are approximately flat rather than loss-making, so FactorVAE's advantage is one of ranking persistence rather than ranking accuracy. Taken together, the findings argue that feature compression should be treated as an empirical hypothesis, validated against the economic objective on the specific universe at hand, rather than adopted as a default preprocessing step on theoretical grounds.

Chapter 2

Related Work

This sector surveys the three bodies of literature that underpin the experimental design of the thesis. Section 2.1 reviews the machine learning literature on cross-sectional return prediction, with particular attention to the role of gradient boosting and learning-to-rank objectives. Section 2.2 covers dimensionality reduction as a feature engineering discipline, tracing the arc from classical linear methods Principal Component Analysis (PCA) and Factor Analysis (FA) to the deep architectures the GKX Autoencoder and FactorVAE that constitute the nonlinear treatments in this thesis, and closes with the question of cross-sectional feature normalisation. Section 2.3 addresses the evaluation frameworks appropriate for financial machine learning, covering the Information Coefficient as a predictive metric, walk-forward validation as the methodological standard for out-of-sample evaluation, and the statistical inference required to compare competing models.

2.1 Machine Learning for Cross-Sectional Return Prediction

From Factor Models to Machine Learning

The cross-sectional prediction of equity returns has historically been framed within the theory of linear factor models. The CAPM of Sharpe (1964) and Lintner (1965) posited that the cross-section of expected returns is fully explained by a single systematic risk factor, the market premium. The three-factor model of Fama & French (1993) extended this framework to include size and value premia, and the subsequent five-factor model (Fama & French, 2015) added profitability and investment factors. Over several decades,the academic literature proliferated alternative factor specifications the so-called 'factor zoo' whose replicability and economic substance Harvey, Liu & Zhu (2016) subjected to systematic multiple-testing scrutiny, finding that only a handful of the hundreds of proposed factors survive. The landmark contribution of Gu, Kelly & Xiu (2020) reframed the question. Rather than deriving factor specifications from theory, they evaluated a broad class of machine learning methods including penalised linear regression (LASSO, Ridge, Elastic Net), regression trees, random forests, gradient boosted trees, and neural networks applied to a set of 94 firm characteristics drawn from the prior academic literature. Their key finding was that nonlinear models generate out-of-sample return forecasts with meaningfully higher predictive R^2 than both linear models and the historical mean benchmark. Gradient boosted trees were among the strongest performers, achieving predictive R^2 values of approximately 0.33% out-of-sample — a modest figure in absolute terms but large relative to the near-zero performance of traditional factor models as predictive devices. Crucially, the predictive gains translated into economically significant long-short portfolio returns after controlling for standard risk factors.

Gradient Boosting in Financial Prediction

Gradient boosting was introduced by Friedman (2001) as a general framework for fitting additive ensembles of weak learners typically shallow decision trees by iteratively minimising a differentiable loss function. The XGBoost implementation of Chen & Guestrin (2016) introduced second-order gradient statistics, column subsampling, and a regularised tree-splitting criterion that dramatically improved both speed and generalisation performance, making gradient boosting the dominant method on structured tabular datasets in competitive machine learning. LightGBM (Ke et al., 2017) further accelerated training through histogram-based splitting and leaf-wise growth, while CatBoost (Prokhorenkova et al., 2018) introduced ordered boosting to address target leakage in the training procedure. The success of gradient boosting on financial data has a structural explanation. Grinsztajn, Oyallon & Varoquaux (2022) demonstrated in a large-scale benchmark that tree-based models consistently outperform deep learning on tabular datasets particularly on small-to-medium samples because they are inherently robust to uninformative features, do not require careful normalisation, and capture irregular, non-smooth decision boundaries that characterise financial signals. Shwartz-Ziv & Armon (2022)

reinforced this conclusion, showing that XGBoost outperforms a range of purpose-built deep learning models for tabular data across benchmark datasets while requiring substantially less hyperparameter tuning. These properties are directly relevant to the present thesis: the ETF universe comprises 60 instruments with daily observations, yielding cross-sectional training windows of tens of thousands of rows a regime in which gradient boosting has a structural advantage over end-to-end deep learning.

Learning-to-Rank for Portfolio Construction

A key methodological development motivating this thesis is the recognition that cross-sectional return prediction is fundamentally a *ranking* problem, not a regression problem. A long-short equity strategy does not require accurate point forecasts of returns; it requires a correct ordering of assets from most to least attractive. Optimising mean squared error — the conventional regression objective penalises all mis-predictions equally regardless of whether they alter the portfolio composition, making it a poorly matched surrogate for portfolio performance. The learning-to-rank literature, developed primarily in the information retrieval community, provides objectives that are directly calibrated to ranking quality. Pairwise approaches (Herbrich et al., 2000; Burges et al., 2005) minimise the probability of reversing the relative ordering of pairs of items,while listwise approaches (Cao et al., 2007) define a loss over the entire ranked list rather than over individual pairs. Poh et al. (2021) applied pairwise and listwise objectives to cross-sectional momentum strategies and documented Sharpe ratio improvements of approximately threefold relative to regression baselines, with the weakest ranking model still outperforming the best regression benchmark. This evidence is the principal motivation for evaluating ranking objectives `lambdarank` for LightGBM and `YetiRank` for CatBoost alongside regression objectives in this thesis.

2.2 Dimensionality Reduction as Feature Engineering

A library of 80 alpha signals derived from price and volume data is high-dimensional, noisy, and highly correlated. The theoretical motivation for compressing such a library into a lower-dimensional representation is rooted in the latent factor hypothesis (Ross, 1976; Chamberlain & Rothschild, 1983): if asset returns are driven by a small number of common factors, then the 80 signals, each capturing some projection of these factors plus idiosyncratic noise, contain substantial redundancy. A dimensionality reduction step that discards this redundancy while preserving the predictive content of the signals should therefore improve the signal-to-noise ratio of the downstream predictor's input features.

Principal Component Analysis

Principal Component Analysis (PCA), attributed to Hotelling (1933) as a formal statistical method building on Pearson (1901), is the canonical technique for linear dimensionality reduction. PCA finds the orthogonal directions of maximum variance in the feature space by computing the eigenvectors of the sample covariance matrix. Retaining the top K eigenvectors—the principal components—yields a K-dimensional projection that captures the maximum possible proportion of the total variance in K dimensions. In financial applications, PCA has a natural interpretation: the leading principal components correspond to the dominant common factors driving co-movement across assets or signals, analogous to the statistical factor structures identified by Connor & Korajczyk (1988) in equity return data. The key limitation of PCA in the present context is its linearity and its variance orientation. PCA maximises explained variance, which is not the same as maximising predictive utility for return forecasting. Components with large variance may be driven by noise—particularly in high-frequency data, where microstructure effects inflate variance while components with modest variance may contain the most predictive signal. Stock & Watson (2002) demonstrated the effectiveness of PCA factors for macroeconomic forecasting, and more recently Kelly, Pruitt & Su (2019) used PCA as a benchmark for latent factor extraction from firm characteristics, providing a point of comparison against which more flexible methods can be evaluated.

Factor Analysis

Factor Analysis (FA) shares with PCA the goal of explaining co-variation among observed variables through a smaller set of latent factors, but it imposes a different model. Under the FA model, each observed variable is decomposed into a linear combination of K common factor plus a variable-specific unique component (communality vs. uniqueness). Unlike PCA, FA explicitly models and separates the unique variance attributable to each variable from the shared variance attributable to common factors. In finance, this distinction is theoretically motivated: the Arbitrage Pricing Theory of Ross (1976) posits exactly this structure common systematic factors driving correlated returns, plus idiosyncratic residuals uncorrelated across assets. FA therefore provides a more faithful representation of the assumed data-generating process than PCA when the goal is to recover

economically meaningful latent factors. Maximum likelihood FA (Jöreskog, 1969) estimates the factor loadings and unique variances jointly by maximising the likelihood under a Gaussian model, and can be complemented by rotation procedures (e.g., Varimax, Promax) to improve interpretability. In practice, the distinction between PCA and FA is often modest when the number of variables is large relative to the number of retained factors, because the communalities approach unity and the unique variances become negligible (Jolliffe, 2002). The empirical comparison between these two methods in this thesis therefore provides evidence on whether the explicit modelling of unique variance provides incremental benefit in the alpha signal context.

The GKX Autoencoder

Gu, Kelly & Xiu (2021) proposed the conditional autoencoder asset pricing model as a nonlinear generalisation of linear factor models. Rather than reconstructing its inputs like a standard autoencoder (Rumelhart, Hinton & Williams, 1986), the model prices realised returns through the bilinear form $r_{i,t} = \beta_{i,t-1}^{\top} f_t + u_{i,t}$, built from two sub-networks. A *beta network* maps each stock's characteristics $z_{i,t-1}$ to a vector of K conditional factor loadings $\beta_{i,t-1}$ through a stack of fully connected hidden layers (each with batch normalisation (Ioffe & Szegedy, 2015) and a ReLU activation) and a linear output layer. A *factor network*—a single linear layer, so that each factor remains interpretable as a portfolio—maps characteristic-managed portfolio returns $x_t = (Z_{t-1}^{\top} Z_{t-1})^{-1} Z_{t-1}^{\top} r_t$ to K latent factors f_t. The predicted return is the inner product $\beta_{i,t-1}^{\top} f_t$, and the network is trained by minimising the mean-squared pricing error against realised returns, with cross-sections processed one date at a time so that all stocks on a given day share the same latent factors. Because the factor exposures are a flexible nonlinear function of characteristics, the model captures a conditional factor structure not attainable by the linear combinations of PCA or FA. Gu, Kelly & Xiu (2021) evaluated it as a standalone conditional asset pricing framework and documented that it explains a larger proportion of the cross-section of returns than linear factor models, with the implied stochastic discount factor attaining out-of-sample Sharpe ratios substantially higher than the PCA benchmark. A critical distinction for this thesis is that the model was designed and evaluated as a self-contained asset pricing model, not as a feature engineering module feeding a separate downstream predictor. Here it is repurposed as an upstream extractor: the conditional betas, and the resulting expected-return signal, are handed to a gradient-boosting ranker—a departure from its original end-to-end use.

FactorVAE

Duan et al. (2022) introduced FactorVAE, a probabilistic dynamic factor model built on the Variational Autoencoder (VAE) of Kingma & Welling (2013). A VAE replaces the deterministic bottleneck of a standard autoencoder with a stochastic latent variable: an encoder outputs the parameters of a posterior distribution over the latent variables, and a decoder maps samples from that distribution back to the modelled quantity. Training maximises the evidence lower bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(z|x)}\big[\log p_\theta(x|z)\big] - D_{\text{KL}}\big(q_\phi(z|x) \,\|\, p(z)\big), \tag{2.1}$$

which balances reconstruction quality against the KL divergence between the posterior $q_\phi(z|x)$ and the prior $p(z)$—a standard Gaussian in the base VAE, and, in FactorVAE, the distribution produced by the learned prior factor predictor. FactorVAE adapts this framework to cross-sectional return prediction through a prior–posterior learning scheme. During training, a posterior factor encoder is allowed to observe future returns and extracts an "optimal" set of latent factors that explain the realised cross-section, while a prior factor predictor—restricted to information available ex ante—is trained to approximate this posterior through the KL term, so that at prediction time the factors can be forecast from observable features alone. A decoder reconstructs the cross-section of stock returns from the latent factors and their loadings, and the probabilistic latent space additionally yields estimates of return variances, giving the model an explicit representation of risk. The model was evaluated on Chinese A-share data and demonstrated improvements over both linear factor benchmarks and standard autoencoder specifications in return predictability and portfolio performance. As with the GKX Autoencoder, however, FactorVAE was designed as an end-to-end model rather than a feature extractor, and its evaluation as an upstream module for gradient boosting is novel to this thesis.

Evidence on Deep Feature Extraction

Beyond the asset pricing literature, the representation learning literature provides broader evidence on the conditions under which deep methods produce useful features. Bengio, Courville & Vincent (2013) surveyed the theoretical and empirical basis for unsupervised pre-training as a regularisation strategy: by learning a compact encoding of the input distribution, a pre-trained encoder provides a better initialisation and more informative features for a supervised downstream task. This view motivates the use of autoencoders and VAEs as feature extractors even in settings where they were not originally designed for that role. In the specific context of stock market prediction, Htun, Biehl & Petkov (2023) surveyed feature selection and extraction techniques

across 32 studies and identified autoencoders and PCA as among the most effective methods for reducing the dimensionality of financial feature spaces. Critically, however, their review did not include a controlled head-to-head comparison of these methods on a unified pipeline—a gap that this thesis directly fills. The absence of such a controlled comparison, combined with the novelty of evaluating the GKX Autoencoder and FactorVAE as feature extractors rather than end-to-end predictors, defines the empirical contribution of this thesis.

Feature Normalisation in the Cross-Section

A design choice that is logically prior to dimensionality reduction, yet frequently left implicit, is the axis along which features are normalised. Two schemes dominate. *Time-series* (standard) normalisation centres and scales each feature by its own historical mean and standard deviation, making values comparable across time for a given asset but preserving the cross-sectional level on any single date. *Cross-sectional* normalisation instead demeans and rescales each feature across all assets at each date, retaining only an asset's standing relative to its peers and removing the common, market-wide component. The cross-sectional scheme is the implicit standard in the machine-learning asset-pricing literature: Gu, Kelly & Xiu (2020, 2021) rank-normalise each characteristic into $[-1, 1]$ within each period, and the construction of factor portfolios more generally relies on cross-sectional standardisation of signals (Asness, Moskowitz & Pedersen, 2013). The rationale is that a dollar-neutral long–short strategy trades relative, not absolute, signal levels, so the time-common component is noise for a rank-based objective. However, cross-sectional demeaning is estimated from the assets available on each date, and in a narrow universe this estimate is itself noisy a tension that is directly relevant to the 60-instrument ETF universe studied here and that motivates the explicit comparison of the two normalisation schemes in this thesis.

2.3 Evaluation Frameworks for Financial Machine Learning

The Information Coefficient

The Information Coefficient (IC) is the canonical performance metric for cross-sectional return prediction in quantitative finance. In its standard form, the IC at time t is the cross-sectional Pearson correlation between predicted returns and realised returns across the N assets in the universe:

$$\mathrm{IC}_t = \mathrm{Corr}\Big(\hat{f}_{i,t},\, r_{i,t+1}\Big), \quad i = 1, \ldots, N, \tag{2.2}$$

where $\hat{f}_{i,t}$ is the return forecast for asset i at time t and $r_{i,t+1}$ is the realised return in the subsequent period. The Rank IC, which uses Spearman rank correlation rather than Pearson, is the appropriate counterpart when the predictive model produces rank outputs rather than point forecasts, and is the primary *predictive* diagnostic throughout this thesis, with the Sharpe ratio of the realised long–short portfolio serving as the primary evaluation criterion Section 3.5. The IC has several properties that make it particularly suitable for cross-sectional prediction evaluation. First, it is scale-invariant: multiplying all forecasts by a constant does not affect the IC, so the evaluation is robust to differences in forecast scaling across methods. Second, it directly captures the ranking quality of the predictor, which is the economically relevant quantity for long-short portfolio construction. Third, the IC framework admits a natural decomposition: the time-series mean IC measures average predictive skill, and the Information Coefficient Information Ratio (ICIR = mean IC / std IC) measures the consistency of that skill.

Walk-Forward Validation

The methodological standard for evaluating predictive models on financial time series is walk-forward validation a form of cross-validation that respects the temporal ordering of the data by ensuring that the model is always trained on past data and evaluated on future data. Unlike k-fold cross-validation, which randomly assigns observations to folds and is therefore inappropriate for time series data where future information must not leak into training, walk-forward validation simulates the actual deployment of the model: at each evaluation date, the model is trained on all data available up to that point and tested on a forward window of previously unseen data. The walk-forward framework can be implemented in two variants. In the *expanding window* variant, the training set grows with each fold as additional data accumulates; in the *rolling window* variant, a fixed-length training window moves forward in time, discarding the oldest observations. In this thesis, a capped expanding window is used: the training set expands but is capped at a maximum of 4 years of historical data, striking a balance between data efficiency and regime relevance. A critical but frequently overlooked requirement of walk-forward evaluation is that all preprocessing transformations must be fitted exclusively on training data within each fold and applied to test data without refitting. This requirement — which applies to standardisation, PCA rotation matrices, factor loadings, and neural network weights alike prevents a subtle

form of information leakage: if a preprocessing step is fitted on the combined train and test data, the test-period statistics will influence the feature transformation applied to training data, and the out-of-sample evaluation will be optimistically biased. Many studies in the financial ML literature omit this precaution, fitting PCA or standardisation globally before splitting into train and test sets a practice that contaminates the out-of-sample evaluation with information from the test period. Bailey et al. (2014) and López de Prado (2018) have documented extensively how backtesting practices in quantitative finance generate optimistically biased performance estimates through multiple channels: in-sample fitting, parameter snooping, and inadequate cross-validation. The strict walk-forward framework implemented in this thesis with preprocessing refitted inside each fold and a temporal embargo between training and test periods addresses these concerns and ensures that reported IC values and portfolio metrics are genuine out-of-sample estimates.

Statistical Inference for Performance Comparison

Point estimates of the Information Coefficient and the Sharpe ratio are not sufficient to establish that one configuration outperforms another; financial return series are short, heavy-tailed, and serially dependent, so apparent differences may reflect sampling variation rather than genuine skill. A rigorous comparison therefore requires formal inference. For predictive accuracy, the statistical significance of a mean Information Coefficient — or of a difference in ICs between a compressed configuration and the baseline — is assessed using Newey & West (1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors, which correct for the serial dependence that pervades daily IC series. The same HAC framework is applied to the market-orthogonality regressions used to establish whether realised performance loads on the regression intercept rather than on residual market exposure. For economic performance, the sampling distribution of the Sharpe ratio is itself non-trivial: Lo (2002) derives its asymptotic standard error and shows how serial correlation distorts naive estimates. Rather than adopt that parametric correction, this thesis takes a distribution-free approach that resamples the return series directly. The stationary block bootstrap of Politis & Romano (1994) preserves the temporal dependence structure by resampling contiguous blocks rather than individual observations, yielding valid inference for Sharpe-ratio differences without parametric assumptions. It is therefore the basis on which each configuration is compared against the baseline on the primary economic criterion. Finally, because several configurations are evaluated against a common baseline, the reported p-values are unadjusted and are therefore interpreted as evidence about individual comparisons rather than as family-wise error rates. Where a family of comparisons is assessed jointly, a Holm–Bonferroni correction is applied and reported alongside the unadjusted values, in the spirit of the warnings of Harvey, Liu & Zhu (2016) and Bailey et al. (2014) against the proliferation of spurious findings in backtested finance. The present thesis therefore rests on two complementary inferential tools — HAC-based tests for Rank IC and for market-orthogonality regressions, and block-bootstrap Sharpe-difference tests for portfolio performance — when assessing whether dimensionality reduction confers a statistically meaningful advantage.

Chapter 3

Methodology

This section details the empirical framework designed to evaluate dimensionality reduction methods as upstream feature extractors for tree-based gradient boosting models.

3.1 Asset Universes and Feature Library

The empirical analysis is conducted across two distinct asset universes to ensure the generalisability of the findings across different market microstructures and regimes. Daily price and volume data are sourced from Yahoo Finance.

- **ETF Universe (2006–2025):** A highly liquid universe of 60 Exchange-Traded Funds spanning diverse equity sectors, geographic regions, fixed income, and commodities. Daily history was collected from 2006, but the out-of-sample evaluation was restricted to 2019–2025 because of the computational cost of the walk-forward procedure, yielding seven annual folds. Since the training window is capped at four years, the folds draw only on data from 2015 onwards (Table 3.1), so the 2008 Global Financial Crisis lies outside both the training and the test samples. The evaluated period nonetheless spans distinct regimes, including the 2020 COVID-19 crash and the 2022 rate shock.
- **S&P 500 Universe (2020–2025):** A large-cap equity universe capturing a structurally distinct, post-pandemic market regime dominated by high cross-sectional volatility and shifting monetary policy. The OOS evaluation was conducted over 2022–2025, yielding four annual out-of-sample folds.

The feature space consists of a curated library of 80 alpha signals derived from the WorldQuant 101 Formulaic Alphas (Kakushadze, 2016), implemented following Jansen (2020). These signals rely exclusively on price and volume data, ensuring standardisation across both asset universes. The raw, wide-format alpha matrices are melted into a longitudinal panel dataset and joined with forward target returns, where the target variable is defined as the return from the close at time $t+1$ to the close at time $t+2$. Execution is modelled using market-on-close orders, assuming complete fills at the official closing price at $t+1$. In aggregate, the out-of-sample panels comprise 105,480 ETF observations (60 instruments across seven annual folds) and 483,966 S&P 500 observations (483 constituents across four annual folds); the S&P 500 count reflects the constituents for which a complete price and alpha history is available over the evaluation window.

3.2 Preprocessing Architecture

The pipeline is structured as a strict walk-forward validation engine to simulate the real-world deployment of a quantitative trading strategy and definitively prevent information leakage. For each validation anchor period of one full year, the system defines a capped historical training window of up to 4 years. Crucially, all data preprocessing—feature normalisation and dimensionality reduction alike—is fitted exclusively on the training folds.

Two alternative normalisation schemes are compared throughout this thesis, and every configuration is run under each. *Cross-sectional* normalisation rescales each feature across all assets at a given time t, so that values are comparable across the cross-section:

$$\tilde{x}_{i,t} = \frac{x_{i,t} - \mu_t}{\sigma_t} \tag{3.1}$$

where μ_t and σ_t denote the cross-sectional mean and standard deviation of the feature across all assets i at time t. *Standard* normalisation instead applies a Z-score along the time dimension, transforming each raw feature

$x_{i,t}$ to have zero mean and unit variance based on the training-period statistics:

$$z_{i,t} = \frac{x_{i,t} - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{3.2}$$

The two schemes are applied as alternatives rather than in sequence: each transforms the raw feature $x_{i,t}$, and the resulting feature sets are carried independently through the remainder of the pipeline. Their relative merits are examined in Section 4.2. These learned transformations, along with the PCA eigenvectors, FA loadings, or neural network weights, are subsequently applied to the out-of-sample test period without refitting. Following predictions, the backtesting module routes the outputs to simulate long-short portfolio construction. The resulting fold schedule is summarised in Table 3.1. Each test year is preceded by a capped training window of up to four prior years; the earliest S&P 500 folds use shorter windows because daily data begin in 2020. Every out-of-sample fold spans a full trading year (≈ 250 days).

Table 3.1: Walk-forward fold schedule. Each row is one annual out-of-sample fold, preceded by a capped (up to four-year) training window. ETF folds use 60 instruments; S&P 500 folds use 483.

Universe	Training window	Test year	Test trading days
ETF	2015–2018	2019	252
ETF	2016–2019	2020	253
ETF	2017–2020	2021	252
ETF	2018–2021	2022	251
ETF	2019–2022	2023	250
ETF	2020–2023	2024	252
ETF	2021–2024	2025	248
S&P 500	2020–2021	2022	251
S&P 500	2020–2022	2023	250
S&P 500	2020–2023	2024	252
S&P 500	2021–2024	2025	249

3.3 Dimensionality Reduction and Predictive Models

The experiment compares five distinct upstream feature configurations, each ultimately evaluated via the out-of-sample Sharpe ratio of the resulting long-short portfolio.

Dimensionality Reduction Configurations:

1. **Baseline:** The full set of 80 standardised raw alphas.
2. **PCA-Only:** Principal Component Analysis, retaining components that maximise linear explained variance.
3. **FA-Only:** Factor Analysis, isolating shared variance from idiosyncratic noise.
4. **AE-Only:** The GKX conditional autoencoder (Gu, Kelly & Xiu, 2021), which prices realised returns through a beta network mapping characteristics to conditional factor loadings and a factor network extracting latent factors from characteristic-managed portfolios.
5. **FactorVAE-Only:** The probabilistic dynamic factor model of Duan et al. (2022), which extracts latent market factors through a prior–posterior scheme trained on the evidence lower bound, with expected returns produced from the prior path at prediction time.

The five configurations are organised into two distinct evaluation tracks. For the **Baseline**, **PCA-Only**, and **FA-Only** configurations, the feature sets are passed directly into the downstream predictive models, with hyperparameter tuning performed jointly across the upstream representation and the predictive model within the same walk-forward procedure.

The **AE-Only** and **FactorVAE-Only** configurations follow a separate two-stage protocol. Each autoencoder architecture is first trained and tuned across its hyperparameter space, with the best configuration selected as the one maximising the mean Sharpe ratio across all walk-forward folds when its latent factors are used as a standalone predictive signal (i.e., without a downstream tree-based model). In this standalone setting, the Sharpe ratio is obtained by ranking stocks according to their expected returns derived directly from the latent representation—specifically, the per-asset return predicted by the model's pricing head, which maps the latent factors (and, for FactorVAE, the inferred factor exposures) to a conditional expected return for each

name on each date—and constructing a long-short portfolio from these rankings. This same best-performing configuration is then also used to generate the latent representations fed into the downstream predictive models, analogous to the PCA and FA representations, with hyperparameter tuning for the tree-based models performed subsequently on top of these fixed latent inputs.

Downstream Predictive Models:
The compressed representations are fed into two tree-based gradient boosting engines, LightGBM and CatBoost. Because the ETF universe comprises 60 instruments, generating cross-sectional training windows of only tens of thousands of observations per fold, deep end-to-end predictors are at a structural disadvantage. As reviewed in Section 2.1, gradient boosting matches or outperforms deep learning on tabular data at this scale, and it imposes no architectural constraint on the upstream feature extractor. It therefore serves as a well-suited, model-agnostic evaluator for the learned features. Models are evaluated using learning-to-rank objectives (`lambdarank` for LightGBM, and `YetiRank` for CatBoost) to align directly with portfolio sorting objectives.

3.4 Hyperparameter Tuning Setup

Hyperparameter optimisation is conducted dynamically within each outer walk-forward training window using the Optuna framework (Akiba et al., 2019) with a Tree-structured Parzen Estimator (TPE) sampler (seed fixed at 42). Because the underlying search spaces for tree models (LightGBM, CatBoost) and feature reduction/neural components (PCA, FA, GKX-AE, FactorVAE) are finite and categorical, pure TPE sampling can revisit identical configurations. To avoid wasting trial budget, a duplicate-control mechanism tracks previously evaluated parameter combinations: if TPE proposes a configuration already seen, the trial is pruned and an unused grid configuration is enqueued in its place. This continues until either the target number of unique completed trials is reached, the grid is exhausted, or an attempt limit is hit. The number of trials per configuration follows a square-root heuristic, $\lceil \sqrt{|G|} \rceil + 10$, capped at 50, where $|G|$ is the grid size, balancing exploration cost against the expense of repeated walk-forward refitting. To respect the temporal ordering of the data, the inner tuning loop applies `TimeSeriesSplit` with three chronological folds derived from the unique trading dates within the current outer training window. An embargo gap is enforced between each training fold and its corresponding validation fold to explicitly mitigate look-ahead bias and serial correlation leakage at the fold boundary. All preprocessing steps, including feature scaling, PCA/FA dimensionality reduction, and neural pricing-equation fitting (GKX-AE or FactorVAE), are fit exclusively on the inner training fold and subsequently applied to the inner validation fold, ensuring no information from the validation or outer test period contaminates the fitted transformations. Crucially, candidate hyperparameter configurations are selected based on their validation Sharpe ratio, computed by running each fold's validation predictions through the same long-short backtesting convention used for out-of-sample evaluation, rather than the native loss function of the underlying learner (e.g., RMSE for regression, or the ranking loss for LambdaRank/YetiRank/pairwise objectives). On financial tabular data, RMSE rapidly converges toward the unconditional standard deviation of asset returns, since the signal-to-noise ratio in return prediction is extremely low; this renders RMSE largely uninformative for distinguishing genuine model skill from noise. The Sharpe-based objective instead ensures that hyperparameters are chosen for their ability to construct a profitable long-short portfolio from the cross-section of assets, directly aligning the validation objective with the economic objective of the final strategy. Validation fold scores are averaged across the three folds to produce the trial's overall objective value, and for neural-feature tree tuning, partial fold scores are reported incrementally to Optuna's median-based pruner (with a startup period of 10 trials and 2 warm-up steps), allowing weak neural configurations to be pruned early after at least two folds have completed, thereby concentrating compute on promising regions of the search space.

Search Spaces

Table 3.2 reports the discrete grid searched for each component. The two linear reducers (PCA, FA) and the two deep extractors (GKX-AE, FactorVAE) share a common latent-dimension grid of $\{5, 8, 10, 12, 15, 20\}$, so that all four methods are compared at matched representation sizes. The resulting grids are moderate in size: $|G| = 648$ for LightGBM and 72 for CatBoost in isolation, rising to 3,888 when a linear reducer is tuned jointly with LightGBM, and 216 and 1,296 for the standalone GKX-AE and FactorVAE searches respectively. Under the $\lceil \sqrt{|G|} \rceil + 10$ heuristic (capped at 50), this yields 36 unique completed trials per fold for LightGBM and 19 for CatBoost in isolation, 50 — the cap — when a linear reducer is tuned jointly with LightGBM, and 25 and 46 for the standalone GKX-AE and FactorVAE searches respectively.

Neural Architecture and Training

Both deep extractors are trained with the Adam optimiser under the learning-rate and weight-decay grids of Table 3.2, for up to 200 epochs with early stopping on the validation objective (patience of 20 epochs for

Table 3.2: Hyperparameter search spaces. Tree models are tuned jointly with the upstream reducer (for PCA/FA) or on top of the fixed best neural representation (for GKX-AE, FactorVAE). Both deep extractors are selected by validation Sharpe ratio.

Component	Hyperparameter	Grid
LightGBM	`learning_rate`	{0.01, 0.05, 0.1}
	`max_depth`	{3, 5, 7}
	`n_estimators`	{300, 500}
	`min_child_samples`	{25, 50, 100}
	`reg_lambda`	{3, 5, 10}
	`subsample`	{0.8, 1.0}
	`colsample_bytree`	{0.8, 1.0}
CatBoost	`learning_rate`	{0.01, 0.05, 0.1}
	`depth`	{4, 6}
	`iterations`	{300, 500}
	`l2_leaf_reg`	{3, 7, 10}
	`colsample_bylevel`	{0.8, 1.0}
PCA / FA	`n_components`	{5, 8, 10, 12, 15, 20}
GKX-AE	`latent_dim`	{5, 8, 10, 12, 15, 20}
	`learning_rate`	$\{1\times10^{-5}, 5\times10^{-5}, 1\times10^{-4}, 5\times10^{-4}\}$
	`weight_decay`	$\{1\times10^{-6}, 1\times10^{-5}, 1\times10^{-4}\}$
	L_1 penalty	$\{0, 1\times10^{-5}, 1\times10^{-4}\}$
	max epochs	200
	early-stop patience	20
FactorVAE	`sequence_length`	{10, 20}
	`hidden_size`	{32, 64, 128}
	`num_factors`	{5, 8, 10, 12, 15, 20}
	`num_portfolios`	{8, 16, 32}
	`learning_rate`	$\{1\times10^{-5}, 5\times10^{-5}, 1\times10^{-4}, 5\times10^{-4}\}$
	`weight_decay`	$\{1\times10^{-6}, 1\times10^{-5}, 1\times10^{-4}\}$
	max epochs	200
	early-stop patience	25

the GKX-AE and 25 for FactorVAE).The GKX conditional autoencoder comprises a beta network that maps the input characteristics through three fully connected hidden layers (64, 48, and 32 units, each with batch normalisation and a ReLU activation) to a linear output layer producing the K conditional factor loadings, and a single-linear-layer factor network mapping characteristic-managed portfolio returns to the K latent factors; it is trained to minimise the mean-squared pricing error on realised returns, with an optional L_1 penalty. FactorVAE is a sequential variational model: it consumes a short look-back window (sequence length of ten or twenty days) through a recurrent feature extractor, encodes it into a stochastic latent of several factors, and is trained on the evidence lower bound of Equation (2.1), with the posterior path used during training and the prior path used to produce expected returns at validation and test time. For both models the best configuration is the one maximising the mean standalone validation Sharpe ratio across folds, after which the latent representation is frozen and passed to the downstream trees.

3.5 Evaluation Metrics

Model performance is evaluated in two stages: predictive accuracy and portfolio economics, with the Sharpe ratio of the simulated long-short portfolio serving as the primary out-of-sample evaluation criterion, supplemented by the Rank IC as a secondary diagnostic of raw predictive skill.

Predictive Diagnostics:

As a secondary, diagnostic measure of predictive skill, the out-of-sample Rank Information Coefficient (Rank IC) is computed daily on a cross-sectional basis for each outer test period. The Rank IC is defined as the Spearman rank correlation between the model's predicted cross-sectional ordering of assets and the realised return ordering, evaluating the monotonic relationship between the predicted and realised ranks. These daily Rank IC values are aggregated across all walk-forward periods, from which the overall Rank IC and the Information Coefficient Information Ratio (ICIR)—the mean Rank IC divided by its standard deviation—are derived. While informative for assessing whether a model's predictions carry any cross-sectional ranking signal at all, the Rank IC is not used as the primary basis for model comparison; final model assessment is instead based on the realised economic performance of the resulting portfolio, as described below.

Economic Metrics:

The primary evaluation criterion is the out-of-sample Sharpe ratio of the simulated long-short portfolio, obtained

by converting raw model signals into target portfolio weights and simulating the resulting book in `vectorbt`. Predictions for each outer test period are pivoted into a date-by-ticker matrix of predicted ranks. On each date, the top-ranked fraction of assets is selected as the long book and the bottom fraction as the short book; the experiments in this thesis use a selection fraction of 5% (top/bottom 5%). The portfolio follows a dollar-neutral long-short construction with 50% gross exposure allocated equally across all selected long positions and 50% gross exposure allocated equally across all selected short positions, giving an equal per-stock weight of $\pm 0.5/k$, where k is the number of names on each side. To respect realistic execution timing, the target weight matrix derived from signals dated t is shifted forward by one trading day and is therefore first executed at the adjusted close of $t+1$, consistent with the supervised target return defined over the $t{+}1 \to t{+}2$ interval. The book is simulated in `vectorbt` with target-percentage weights, shared cash across the long and short legs, an initial capital of \$10,000,000, and universe-specific transaction costs and slippage (0.0001 / 0.0005 for the S&P 500 universe and 0.0001 / 0.0002 for the ETF universe). From the resulting daily portfolio time series, the Sharpe ratio is computed as the headline metric for ranking experiments, alongside Annualised Return, Annualised Volatility, Sortino Ratio, Maximum Drawdown, Calmar Ratio, gross and net exposure, and turnover. Daily turnover is measured from the executed target-weight matrix as $T_t = \sum_i |w_{i,t} - w_{i,t-1}|$, the total notional traded on day t as a fraction of gross exposure, so that a complete replacement of both legs corresponds to $T_t = 2$; the first day of each out-of-sample series, on which the book is built from cash, is excluded. The reported T is the mean of T_t over the out-of-sample period and enters the cost calculation of Section 4.3 directly. Let r_t denote the daily portfolio return and $V_t = \prod_{s \le t}(1 + r_s)$ the cumulative portfolio value. With a risk-free rate of zero and an annualisation factor of 365 calendar days — the convention applied by the `vectorbt` backtester, which infers a calendar-daily frequency from the return index — the reported metrics are defined as

$$\text{Sharpe} = \sqrt{365}\,\frac{\bar{r}}{\sigma_r}, \qquad \text{Sortino} = \sqrt{365}\,\frac{\bar{r}}{\sigma_r^-}, \tag{3.3}$$

$$\text{MDD} = \max_t \left(1 - \frac{V_t}{\max_{s \le t} V_s}\right), \qquad \text{Calmar} = \frac{R_{\text{ann}}}{\text{MDD}}, \tag{3.4}$$

where $\bar{r}$ and σ_r are the mean and standard deviation of r_t, σ_r^- is the downside deviation computed over negative returns only, R_{ann} is the annualised return, and MDD is the maximum drawdown. Note that this calendar-day convention scales all reported Sharpe and Sortino ratios by a factor of $\sqrt{365/252} \approx 1.204$ relative to the trading-day convention. Because every ratio reported here is negative, this magnifies them in absolute value: the trading-day equivalents are uniformly closer to zero, so the figures below are conservative rather than flattering. Since every configuration is annualised identically, the comparisons and significance tests reported below are unaffected. The Information Coefficient Information Ratio is defined analogously over the daily Rank IC series, $\text{ICIR} = \overline{\text{IC}}/\sigma_{\text{IC}}$.

3.6 Statistical Testing Framework

To assess whether differences in predictive and economic performance are statistically meaningful, two complementary families of tests are employed: predictive signal tests based on out-of-sample Rank Information Coefficients (Rank ICs) and portfolio performance tests based on long–short portfolio returns.

Predictive signal tests. For each model configuration, the full time series of daily out-of-sample Rank ICs is analysed. The mean Rank IC and the Information Coefficient Information Ratio (ICIR) are reported as summary measures of predictive performance. Statistical significance is assessed using a two-sided hypothesis test of the mean Rank IC, with inference based on Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors to account for potential serial dependence in the daily IC series. To evaluate whether dimensionality reduction methods significantly alter predictive performance, each compressed configuration is compared directly against the baseline model: daily Rank IC series are aligned on a common date index and transformed into a series of pairwise differences, and the same HAC inference framework is applied to test whether the mean difference is distinguishable from zero.

Portfolio performance tests. Economic performance is evaluated using the daily returns of the corresponding long–short portfolios over the full out-of-sample period. Differences in annualised Sharpe ratios between the baseline model and each alternative configuration are evaluated using a stationary block bootstrap (Politis & Romano, 1994) with 5,000 resamples. By resampling contiguous blocks of trading days rather than individual observations, the bootstrap preserves the temporal dependence structure of financial return series and provides a robust, distribution-free assessment of performance differences.

Multiple comparisons. Because several baseline-versus-alternative comparisons are conducted for each performance metric, reported p-values are unadjusted and are not treated as family-wise error rates. Where a family of comparisons is assessed jointly, a Holm–Bonferroni correction across that family is reported alongside

the unadjusted values in Section 4.3. This framework is applied consistently across both asset universes, with all comparisons restricted to identical prediction tasks, scaling procedures, and out-of-sample evaluation periods.

Chapter 4

Experiments and Results

This section reports the out-of-sample results of the five feature configurations. Section 4.1 presents the headline predictive and economic performance across both universes and both downstream learners, Section 4.2 examines the choice of feature normalisation, and Section 4.3 establishes that the realised performance is orthogonal to the market and characterises how it varies across regimes. The interpretation of these results, and the conclusions that follow from them , are set out in Section 5.

4.1 Out-of-Sample Results

The central finding of this thesis is that dimensionality reduction does not behave as a free lunch in the alpha signal setting. The latent factor hypothesis motivates compression on the grounds that a large library of price- and volume-derived signals is noisy, correlated, and redundant, and that discarding this redundancy should sharpen the predictive content fed to the downstream model. The empirical evidence assembled here qualifies that expectation. Compression delivers its promised benefit only when the reduction step preserves the cross-sectional ranking information that the long–short strategy actually exploits; when it does not, the loss of signal outweighs the reduction in noise, and the Baseline of raw standardised alphas remains a demanding benchmark. The gradient boosting models prove robust to the uninformative and correlated features that the latent factor argument treats as a liability, which narrows the room for an upstream extractor to add value.

Tables 4.1 and 4.2 report the headline out-of-sample results for the two universes under both normalisation schemes. Each block of rows corresponds to one of the five feature configurations evaluated with the two downstream models, and the columns report the out-of-sample Rank IC — the diagnostic of raw predictive skill — and the out-of-sample Sharpe ratio of the realised long–short portfolio — the primary economic criterion — separately for standard and cross-sectional normalisation. Both schemes were run for every configuration, including the deep extractors, so the two columns generally differ. The role of the normalisation choice is discussed further in Section 4.2.

Table 4.1: Out-of-sample results, ETF universe (2019–2025): Rank IC and annualised Sharpe ratio of the long–short portfolio for each configuration and downstream model, under standard and cross-sectional normalisation.

		Standard		Cross-sectional	
Configuration	Model	Rank IC	Sharpe	Rank IC	Sharpe
Baseline	LightGBM	+0.0034	−0.648	+0.0029	−1.333
	CatBoost	+0.0128	−0.734	+0.0158	−1.369
PCA	LightGBM	+0.0014	−1.451	−0.0023	−1.801
	CatBoost	+0.0031	−1.740	+0.0153	−1.077
FA	LightGBM	−0.0027	−2.142	−0.0024	−2.630
	CatBoost	+0.0057	−1.912	+0.0097	−1.322
GKX-AE	LightGBM	+0.0112	−1.981	−0.0003	−1.661
	CatBoost	+0.0040	−1.901	+0.0073	−1.765
FactorVAE	LightGBM	+0.0049	**−0.537**	−0.0008	−0.641
	CatBoost	+0.0119	−0.906	+0.0099	−1.420

Read together, the two tables tell a consistent but sobering story. The out-of-sample Sharpe ratios are negative throughout: under realistic transaction costs and strict walk-forward evaluation, none of the five configurations produces a profitable long–short portfolio over the test periods, and the discussion below is therefore concerned with relative rather than absolute performance. Within that relative comparison, two patterns recur across both universes. First, the FactorVAE representation paired with LightGBM is the least

Table 4.2: Out-of-sample results, S&P 500 universe (2022–2025): Rank IC and annualised Sharpe ratio of the long–short portfolio for each configuration and downstream model, under standard and cross-sectional normalisation.

		Standard		Cross-sectional	
Configuration	Model	Rank IC	Sharpe	Rank IC	Sharpe
Baseline	LightGBM	+0.0061	−1.375	+0.0044	−2.120
	CatBoost	+0.0018	−5.071	+0.0038	−4.605
PCA	LightGBM	+0.0049	−4.331	+0.0048	−3.873
	CatBoost	+0.0021	−3.845	+0.0034	−3.717
FA	LightGBM	+0.0013	−4.460	+0.0034	−4.524
	CatBoost	−0.0010	−4.866	+0.0015	−3.920
GKX-AE	LightGBM	+0.0082	−3.994	+0.0017	−5.121
	CatBoost	+0.0022	−5.386	−0.0003	−6.510
FactorVAE	LightGBM	+0.0032	−1.296	**+0.0066**	**−0.742**
	CatBoost	−0.0007	−2.456	−0.0001	−2.485

unprofitable configuration in each universe, with a Sharpe ratio of −0.537 on the ETF universe (under standard normalisation) and −0.742 on the S&P 500 (under cross-sectional normalisation), and, under cross-sectional normalisation, it attains the highest Rank IC of any configuration on the S&P 500 under that scheme (+0.0066). That its best ETF result comes under standard rather than cross-sectional normalisation is consistent with the narrow-universe effect documented in Section 4.2. This is the closest the study comes to a candidate for the best feature extractor, and it suggests that the disentangled latent factors of the variational model carry ranking information that a ranking-calibrated tree can exploit better than the raw alphas can.The turnover analysis of Section 4.3 qualifies this reading: the advantage rests more on the temporal stability of the ordering the variational model induces than on its point-in-time accuracy. Second, the GKX Autoencoder is consistently among the weakest configurations — it records the worst Sharpe ratio in the S&P 500 universe (−6.510 under CatBoost) — which is consistent with the small-sample argument advanced in the methodology: a deterministic reconstruction objective overfits the limited cross-sections more readily than either the regularised variational encoder or the simple linear projections.

The tables also expose why predictive accuracy cannot be read as a proxy for economic value. The configuration with the highest Rank IC is rarely the one with the best Sharpe ratio. On the ETF universe, CatBoost lifts the Rank IC of every method into clearly positive territory — the Baseline reaches +0.0158 — yet its Sharpe ratios remain worse than those of LightGBM, whose IC values hover near zero. A model can therefore order the bulk of the cross-section slightly better while still mismanaging the extreme long and short positions that determine portfolio returns. This divergence is the empirical justification for selecting models on realised Sharpe rather than on a predictive loss, and it cautions against the common practice of declaring a feature extractor superior on the strength of its IC alone.

Finally, the raw Baseline is not dominated. It posts the strongest ETF Rank IC under CatBoost and competitive Sharpe ratios under LightGBM in both universes, confirming that gradient boosting already absorbs the redundancy in the eighty-signal library that compression is meant to remove. The one exception is FactorVAE, which under LightGBM improves on the no-reduction benchmark on the primary Sharpe criterion in both universes— significantly so on the S&P 500 ($\Delta\text{SR} = +1.38$, block-bootstrap $p = 0.008$), though not on the ETF universe (+0.11, $p = 0.79$). That edge is universe-dependent rather than model-dependent: under CatBoost the Baseline reclaims the ETF universe ($\Delta\text{SR} = -0.17$, $p = 0.78$), whereas on the S&P 500 FactorVAE improves on the Baseline under both learners (+2.12, $p = 0.002$). Nor does the advantage hold uniformly on Rank IC. The sharpest expression of the no-free-lunch conclusion is therefore not that nothing beats the Baseline, but that no upstream method beats it consistently across both downstream learners and both evaluation criteria.

A closer reading of the two tables also separates the two downstream models. CatBoost is the stronger ranker on the ETF universe, where it pushes the Rank IC of every configuration into positive territory and roughly quintuples the IC of the Baseline relative to LightGBM (+0.0158 against +0.0029), yet this ranking advantage does not survive the move to portfolio returns: LightGBM delivers the better Sharpe ratio for three of the five configurations. The same economic pattern holds on the S&P 500 — LightGBM again delivers the better Sharpe ratio for three of the five configurations, by margins as large as the gap between −2.120 and −4.605 for the Baseline — but there CatBoost's ranking advantage does not arise in the first place, the two models being closely matched on Rank IC. The practical reading is that CatBoost's ordered boosting extracts marginally cleaner cross-sectional rankings, but LightGBM's leaf-wise growth produces portfolios whose extreme positions are better behaved, and it is the latter property that the long–short Sharpe ratio rewards. Because the two models disagree about which configuration is best, the conclusions of this thesis are reported at the level of the feature representation rather than tied to a single learner.

Within the family of reduction methods, the results do not support a straightforward ordering in which architectural sophistication translates into better features. The linear methods, PCA and Factor Analysis, provide a competitive and computationally inexpensive form of compression whose behaviour is stable across folds. The two deep generative extractors diverge sharply, however: FactorVAE is the relative leader on the primary Sharpe criterion in both universes, whereas the GKX Autoencoder is among the weakest, so representational flexibility translates into downstream value only when it is regularised. The structural explanation advanced in the methodology is borne out by the evidence: the ETF universe yields cross-sectional training windows of only tens of thousands of observations per fold, a small-sample regime in which deep encoders are at a disadvantage and in which their learned representations are more sensitive to the idiosyncrasies of each training window than the linear projections are.

A second finding concerns the importance of carrying the analysis through to portfolio economics rather than stopping at predictive accuracy. Differences that appear at the level of raw predictive skill do not map cleanly onto differences in risk-adjusted portfolio performance, because the long–short strategy depends on the ordering of assets at the extremes of the cross-section rather than on the average quality of the forecast. This divergence reinforces the methodological decision to select models on the Sharpe ratio of the realised portfolio rather than on a regression loss, and it cautions against inferring economic value from predictive metrics alone.

Finally, the comparison across the two universes shows that conclusions about feature extraction are regime-dependent. The ETF universe, spanning multiple market cycles, and the post-pandemic S&P 500 universe, characterised by high cross-sectional volatility and shifting monetary policy, do not award the same ranking to the five configurations. No single dimensionality reduction method dominates across asset classes and market environments, which argues against treating any one extractor as a universal default and in favour of selecting the reduction stage with reference to the characteristics of the target universe.

4.2 The Effect of Feature Normalisation

A factor that operates beneath the five configurations, and that materially shapes the headline results, is the choice of feature normalisation. The two schemes compared in this thesis differ in the axis along which each alpha signal is rescaled. *Standard* normalisation applies a time-series z-score: each feature is centred and scaled by its own mean and standard deviation estimated on the training window, so its values are made comparable across time for a given asset, but the cross-sectional level of the feature on any single day — the common, market-wide component that moves all assets together — is left intact. *Cross-sectional* normalisation instead demeans and rescales each feature across all assets at each date, so that what survives is each asset's relative standing within that day's cross-section and the common component is removed.

The distinction matters because the prediction target of this thesis is a *relative* one. Rank IC measures the quality of the cross-sectional ordering, and the dollar-neutral long–short portfolio trades the difference between the top and bottom of that ordering; the absolute level of a signal on a given day carries no information that the strategy can monetise. Cross-sectional normalisation aligns the feature space with this objective by stripping out the time-common variation — broad market or sector moves — that is noise for a rank-based task, and by equalising the feature scale within each daily cross-section that the model must order. The full per-configuration figures under both schemes are collected in Tables 4.1 and 4.2; the discussion here reads those tables along the normalisation axis. The deep extractors were also run under both normalisation schemes in the tree-fed pipeline, so their standard and cross-sectional entries in Tables 4.1 and 4.2 generally differ, just as they do for the linear configurations. The standalone versions of the same extractors, reported in Table 4.3 below, were likewise run under both schemes and respond to the choice. Reading down the linear configurations — Baseline, PCA, and Factor Analysis — a pattern emerges on the predictive metric, but it is conditional on the downstream learner: cross-sectional normalisation raises the Rank IC in eight of the twelve comparisons, and does so in all six under CatBoost against only two of the six under LightGBM. The effect is clearest under CatBoost, where the PCA representation on the ETF universe improves from $+0.0031$ to $+0.0153$ and Factor Analysis from $+0.0057$ to $+0.0097$, and where Factor Analysis on the S&P 500 moves from a negative -0.0010 to a positive $+0.0015$. This is the expected consequence of removing the time-common component: once each feature is judged only by an asset's standing relative to its peers on the same day, the signal that a rank-based learner can exploit becomes cleaner.

The Sharpe ratio, however, does not follow the Rank IC uniformly, and the direction of the effect depends on the universe. On the S&P 500, cross-sectional normalisation improves risk-adjusted performance for the linear configurations under CatBoost, where all three gain — PCA improves from -3.845 to -3.717 and Factor Analysis from -4.866 to -3.920 — although across all ten S&P 500 comparisons the effect is evenly split. Where it helps, it does so because that test period is dominated by large, rate-driven market-wide swings, and demeaning each day removes precisely this common beta, leaving the dollar-neutral book cleaner. On the ETF universe the effect on the Sharpe ratio is frequently the opposite: the Baseline deteriorates from -0.648 to -1.333 under LightGBM, and Factor Analysis from -2.142 to -2.630. The explanation is the breadth of the

universe. With only sixty instruments, the daily cross-sectional mean and standard deviation are estimated from very few names, so subtracting a noisily estimated common component injects estimation error into every feature and can erode the position-sizing information that the standard scheme retains, even as the average ranking improves. Feature normalisation is therefore not a neutral preprocessing detail: it improves raw ranking skill consistently but converts that skill into better portfolio economics only when the cross-section is wide enough and the common component large enough for its removal to help. It is one more channel through which a gain in predictive IC fails to guarantee a gain in realised performance.

A complementary question is how the deep extractors behave as *standalone* predictors — ranking assets directly on the latent expected-return signal of the encoder, with no downstream tree. Table 4.3 reports these runs, which were available under both normalisation schemes, and two observations stand out. First, on the ETF universe the standalone FactorVAE under standard normalisation produces the single best risk-adjusted result anywhere in this study, a Sharpe ratio of -0.248, together with the highest Rank IC of the standalone runs ($+0.0117$), and the standalone GKX Autoencoder is likewise stronger here than its tree-fed counterpart. Used directly, the encoders' own ordering of the cross-section is more informative than what a tree recovers after the latent factors have been handed to it, which suggests that on a narrow universe the tree stage can dilute rather than refine an already-compact deep representation. Second, the normalisation preference inverts relative to the tree-fed pipeline: for the standalone signal, standard normalisation is clearly better on the ETF universe — on both Rank IC and Sharpe — whereas cross-sectional normalisation again helps the FactorVAE Sharpe on the larger, beta-dominated S&P 500 (-1.049 against -1.833).

Table 4.3: Standalone deep extractors, without a downstream tree model. The long–short portfolio is formed by ranking assets directly on the latent expected-return signal of the GKX Autoencoder or FactorVAE. Out-of-sample Rank IC and Sharpe ratio under both normalisation schemes.

		Standard		Cross-sectional	
Universe	Extractor	Rank IC	Sharpe	Rank IC	Sharpe
ETF	FactorVAE	$+0.0117$	$\mathbf{-0.248}$	$+0.0057$	-0.944
	GKX-AE	$+0.0087$	-0.995	$+0.0035$	-1.854
S&P 500	FactorVAE	-0.0007	-1.833	-0.0017	-1.049
	GKX-AE	-0.0018	-4.225	-0.0030	-5.227

The reason for the inversion is that an expected-return signal, unlike a pure ranking, retains useful information in the cross-sectional *level* of the latent factor: standard normalisation preserves that level, while cross-sectional demeaning discards it, which is costly precisely when the cross-section is narrow and the daily mean is poorly estimated. The standalone results therefore reinforce the central message of this section — the right normalisation is not a fixed default but depends jointly on the breadth of the universe and on whether the signal is consumed as a ranking or as a level — and they add a practical caveat: on a small universe, a well-regularised deep extractor may be more useful read directly than passed through an additional tree stage.

4.3 Robustness: Market Orthogonality and Regime Conditioning

The headline result of this thesis is uncomfortable — every long–short configuration loses money out of sample — and a careful reader is entitled to ask whether the losses are an artifact of hidden market exposure rather than a property of the signals themselves. A dollar-neutral book that is inadvertently net short would lose in a rising market for reasons that have nothing to do with feature quality. This subsection rules that explanation out and, in doing so, characterises *how* the strategies fail. The market proxy is the equal-weight daily return of the traded universe, reconstructed from the realised per-asset returns, and inference uses Newey–West HAC standard errors.

For each configuration the realised daily long–short return is regressed on the contemporaneous universe return,

$$r_t^{\mathrm{LS}} = \alpha + \beta_{\mathrm{mkt}}\, r_t^{\mathrm{mkt}} + \varepsilon_t, \tag{4.1}$$

and the estimated market beta, annualised alpha, their HAC t-statistics, the regression R^2 and the return correlation are reported in Tables 4.4 and 4.5 for the LightGBM configurations under each universe's more favourable normalisation (standard for the ETFs, cross-sectional for the S&P 500; CatBoost and the alternative scheme are quantitatively similar). Annualised alphas are reported as $252\,\hat{\alpha}$, on the trading-day basis used for the cost drag below, so that the two are directly comparable; this differs from the 365-calendar-day convention applied to the Sharpe ratios in Section 3.5.

Two facts stand out. First, the strategies are genuinely market-orthogonal: every market beta is economically negligible (all $|\beta_{\mathrm{mkt}}| \le 0.04$ in the reported specifications), the return correlation with the universe never exceeds 0.07 in absolute value, and the regression R^2 stays below half a percent throughout. Not a single beta

Table 4.4: Market-orthogonality regression (4.1), ETF universe (LightGBM, standard normalisation). Betas are economically negligible and the regression R^2 is near zero; the negative annualised alphas are statistically significant under Newey–West inference, except for FactorVAE and the Baseline.

Config	β_{mkt}	$t(\beta)$	α_{ann} (%)	$t(\alpha)$	R^2	Corr
Baseline	+0.033	+1.85	−5.54	−1.68	0.0037	+0.061
PCA	−0.015	−0.75	−9.02	−3.30	0.0012	−0.034
FA	+0.005	+0.32	−14.18	−4.86	0.0001	+0.011
GKX-AE	+0.011	+0.75	−11.93	−4.60	0.0007	+0.026
FactorVAE	+0.021	+1.08	−4.18	−1.36	0.0017	+0.042

Table 4.5: Market-orthogonality regression (4.1), S&P 500 universe (LightGBM, cross-sectional). FactorVAE is the only configuration whose annualised alpha is not statistically distinguishable from zero; the remaining configurations carry large, significantly negative alpha at negligible beta.

Config	β_{mkt}	$t(\beta)$	α_{ann} (%)	$t(\alpha)$	R^2	Corr
Baseline	+0.023	+1.24	−15.23	−4.02	0.0021	+0.046
PCA	+0.004	+0.33	−17.11	−6.40	0.0002	+0.013
FA	−0.002	−0.21	−21.16	−8.24	0.0000	−0.007
GKX-AE	+0.016	+1.39	−25.20	−8.64	0.0022	+0.047
FactorVAE	+0.005	+0.35	−5.45	−1.30	0.0001	+0.011

is statistically significant at the five-percent level — the largest, the ETF Baseline, reaches only $t = 1.85$ — so whatever small loadings appear are too small to move portfolio returns materially. The dollar-neutral construction does what it is designed to do, and the losses cannot be blamed on a directional market bet. Second, and consequently, the negative performance loads almost entirely onto the regression intercept: the annualised alphas are negative and, on the S&P 500, statistically significant for every configuration except FactorVAE; on the ETF universe the Baseline ($t = -1.68$) is likewise not distinguishable from zero at the five-percent level. The strategies are not unlucky beta: whatever destroys value does so orthogonally to the market.The intercept, however, absorbs two economically distinct effects — transaction-cost drag and genuine selection failure — since both are by construction uncorrelated with the market return and therefore load on α rather than on β_{mkt}. Writing c for the per-unit trading cost and T for mean daily turnover, the implied annualised drag is $252\,T\,c$. Realised turnover, measured from the executed weight matrices as defined in Section 3.5, is reported in Table 4.6. It is high: averaged across all forty-eight configurations $T = 1.46$, so roughly one and a half times the gross book is traded every day, with individual configurations spanning $T = 0.71$ to $T = 1.78$. Applied across that range, $c = 6$ basis points on the S&P 500 implies a drag of 10.7% to 26.9% per annum, and $c = 3$ basis points on the ETF universe, of 5.4% to 13.5%.

These figures are of the same order as the estimated intercepts themselves, and in most cases larger. Adding the drag back to the annualised alphas of Table 4.5 gives implied gross alphas of +6.8% for the Baseline, +6.3% for PCA, +3.5% for Factor Analysis, −0.2% for the GKX Autoencoder and +8.5% for FactorVAE; the corresponding ETF figures are +4.8%, +3.4%, −1.6%, +1.5% and +3.8%. Gross of costs, in other words, the strategies are approximately flat rather than systematically perverse. The negative net alphas documented above are not evidence of inverted selection or of a misaligned target; they are evidence of a signal too weak to pay for the turnover that this portfolio construction rule demands. This is the sharpened form of the no-free-lunch conclusion: the failure lies not in the sign of the signals but in the ratio of their gross edge to the cost of trading them.

The decomposition also relocates FactorVAE's advantage. It is the only low-turnover family in the study (T between 0.71 and 1.09, against 1.27 to 1.78 for every other feature set), and this accounts for most of its edge. Of its 9.7 percentage-point net-alpha advantage over the Baseline on the S&P 500, 8.0 points are attributable to lower cost drag and only 1.7 to a higher gross alpha; on the ETF universe the entire 1.4-point advantage comes from turnover, and FactorVAE's implied gross alpha is in fact one point *below* the Baseline's. The variational representation therefore produces a more *persistent* cross-sectional ordering rather than a more accurate one — which is precisely why its economic advantage was not accompanied by any detectable gain in Rank IC (Section 4.1). The regularisation that distinguishes it from the deterministic autoencoder buys stability of the ranking through time, and at these cost levels stability is worth more than marginal accuracy.

FactorVAE is the consistent exception. Its annualised alpha is the least negative in both universes (−4.18% on the ETFs, −5.45% on the S&P 500) and is the only alpha that a Newey–West test cannot distinguish from zero in *both* universes ($t = -1.36$ and $t = -1.30$). In risk-adjusted terms it is the only representation that is not destroying value at a statistically detectable rate, which corroborates, on an orthogonality-adjusted basis, the ranking already visible in the Sharpe ratios of Tables 4.1 and 4.2.A direct test of the difference against the

Table 4.6: Realised mean daily turnover $T = \overline{\sum_i |w_{i,t} - w_{i,t-1}|}$ and implied annualised cost drag $252\,T\,c$, measured from the executed weight matrices. $c = 3$ bp on the ETF universe and 6 bp on the S&P 500. LightGBM configurations under each universe's more favourable normalisation; CatBoost and the alternative scheme are quantitatively similar. FactorVAE is the only low-turnover representation.

	ETF (standard)		S&P 500 (cross-sec.)	
Config	T	Drag (%)	T	Drag (%)
Baseline	1.36	10.3	1.45	22.0
PCA	1.64	12.4	1.55	23.4
FA	1.67	12.6	1.63	24.7
GKX-AE	1.77	13.4	1.66	25.0
FactorVAE	1.06	8.0	0.93	14.0

Baseline confirms this ranking and locates its limits. Comparing annualised Sharpe ratios with a stationary block bootstrap (5,000 resamples, mean block length 21 trading days), FactorVAE's improvement over the raw-alpha Baseline under LightGBM is statistically significant on the S&P 500 (ΔSR $= +1.38$, $p = 0.008$, 95% CI $[+0.27, +2.44]$; the p-value remains below 0.01 for mean block lengths from five to sixty-three days) but not on the ETF universe (ΔSR $= +0.11$, $p = 0.79$), where the two configurations are statistically indistinguishable. The same test applied to the remaining configurations sharpens the no-free-lunch conclusion considerably: on the S&P 500 under LightGBM, PCA ($p = 0.007$), Factor Analysis ($p < 0.001$) and the GKX Autoencoder ($p < 0.001$) are all significantly *worse* than the uncompressed Baseline. Because four alternatives are compared against a common baseline on this universe, a Holm–Bonferroni correction was applied across the family: tested in ascending order against thresholds of 0.0125, 0.0167, 0.025 and 0.05, all four comparisons (Factor Analysis and the GKX Autoencoder at $p < 0.001$, PCA at $p = 0.007$, FactorVAE at $p = 0.008$) remain significant at the five-percent level, so the conclusions are not artefacts of multiple testing. A Newey–West test on the paired daily Rank IC differences, by contrast, is insignificant throughout ($p = 0.74$ on the ETFs, $p = 0.49$ on the S&P 500 for FactorVAE), so the economic advantage is not accompanied by a detectable gain in ranking accuracy — and the GKX Autoencoder illustrates the divergence most starkly, recording the most significant Rank IC of any ETF configuration ($+0.0112$, $t = 2.90$, $p = 0.004$) while simultaneously destroying 1.33 units of Sharpe ratio relative to the Baseline ($p = 0.027$). The market-neutrality of the book is shown directly in Figure 4.1: the cloud of daily returns is flat against the universe return, with a fitted slope indistinguishable from zero.

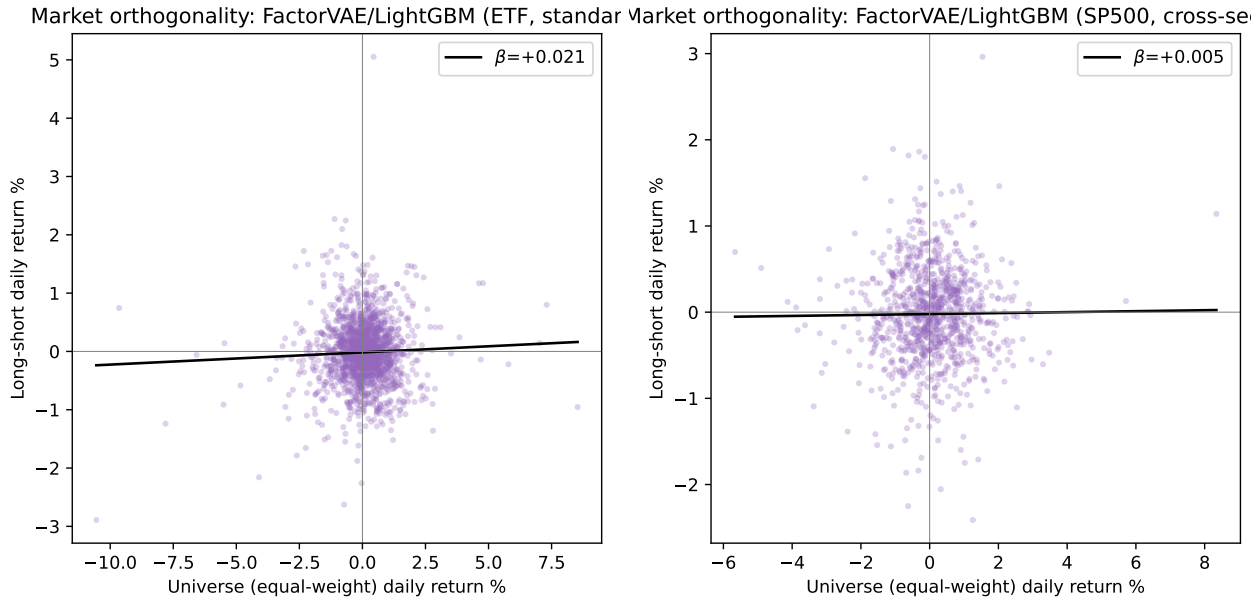


Figure 4.1: Daily long–short return against the equal-weight universe return for the FactorVAE/LightGBM book (ETF standard, left; S&P 500 cross-sectional, right). The near-flat regression line ($\beta \approx 0$) visualises the market orthogonality: portfolio returns carry essentially no linear exposure to the broad universe move.

The temporal structure of the losses is shown in Figures 4.2, 4.3 and 4.4, and quantified by year in Table 4.7. Two regularities tie the economic picture back to the orthogonality result. The 2022 rate shock — the worst single year of the out-of-sample S&P 500 period — punished every configuration roughly equally (annual returns between -19% and -38%), which is exactly what a market-neutral book exposed to a broad, cross-sectionally

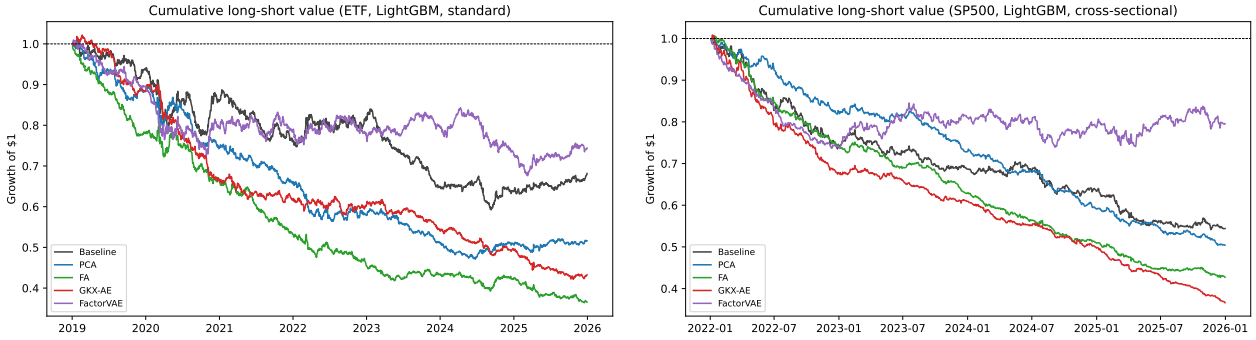


Figure 4.2: Cumulative growth of $1 for the dollar-neutral long–short book (LightGBM), each universe under its more favourable normalisation: ETF standard (left) and S&P 500 cross-sectional (right). FactorVAE is the only representation whose equity curve stabilises rather than bleeding monotonically; the divergence is starkest on the S&P 500, where FactorVAE holds near 0.8 after the 2022 drawdown while the other configurations decline to between 0.4 and 0.55.

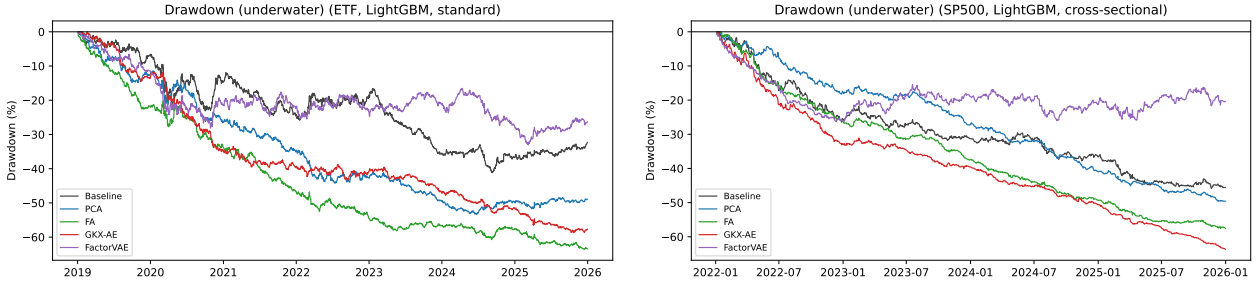


Figure 4.3: Underwater (drawdown) curves for the same books, ETF (left) and S&P 500 (right). The depth and persistence of the drawdowns make the relative robustness of FactorVAE visible: its book spends far less time deep underwater than the linear and deterministic-autoencoder alternatives.

indiscriminate repricing should experience. Notably, FactorVAE offers no protection in the stress year itself: its 2022 loss of −29.8% is marginally larger than the Baseline's −28.7%.Its advantage is concentrated entirely in the calmer post-2022 regime, where the configurations separate and FactorVAE is the only one to post positive calendar years (+9.5% in 2023 and +1.8% in 2025), while the linear and deterministic-autoencoder representations continue to bleed . Conditioning daily returns on the sign of the market return tells the same story from a different angle: mean long–short returns are similar — and similarly negative — on up-market and down-market days, confirming that the strategies neither systematically profit from nor hedge against market direction.

Table 4.7: Annual long–short return (%) by calendar year, S&P 500 universe (LightGBM, cross-sectional). The 2022 rate shock penalises all configurations; only FactorVAE recovers to positive territory in the calmer subsequent years.

Config	2022	2023	2024	2025
Baseline	−28.7	−8.0	−7.9	−14.8
PCA	−18.8	−12.7	−20.7	−15.7
FA	−29.3	−16.9	−20.8	−17.4
GKX-AE	−38.4	−10.6	−20.2	−30.5
FactorVAE	−29.8	+9.5	−2.8	+1.8

Taken together, the orthogonality and regime evidence upgrades the central finding from a bare statement that the strategies lose money to a precise characterisation of why: the losses are market-neutral, concentrated in a single stress regime, and meaningfully smaller — to the point of statistical insignificance — only for the regularised variational representation. This is the form in which the result is most useful to a practitioner, because it locates the failure in the signal-to-portfolio mapping rather than in incidental market exposure — and, within that mapping, attributes the bulk of the shortfall to the cost of the daily rebalancing rule rather than to the ordering the signals produce.

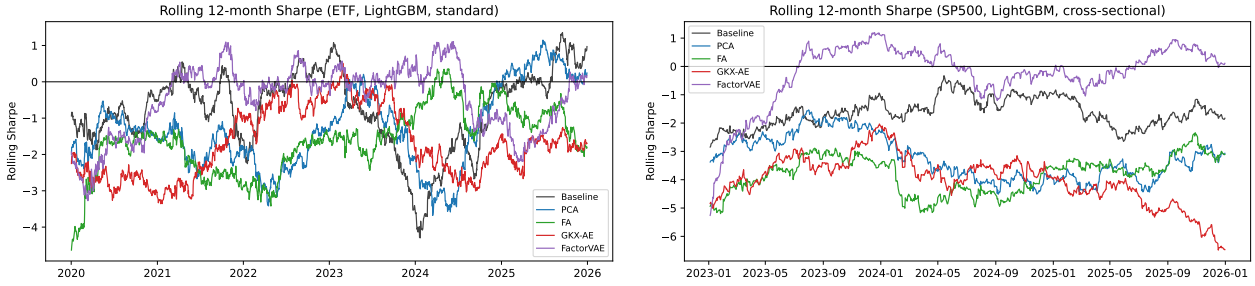


Figure 4.4: Rolling 12-month annualised Sharpe ratio, ETF (left) and S&P 500 (right). Skill is strongly time-varying for every configuration; FactorVAE is the only one to spend sustained periods in positive territory, concentrated in the calmer post-2022 sub-sample.

Chapter 5

Discussion and Future Work

This section interprets the results reported in Section 4. It first answers the three research questions posed in Section 1.2, then sets out the contributions of the work and its practical implications, the limitations that bound its conclusions, and the directions for further research that it opens.

5.1 Answers to the Research Questions

Each question can now be answered in turn.

Q1: Does compression improve on the raw alpha library? Not reliably. Under LightGBM, FactorVAE is the only representation to improve on the raw-alpha Baseline on the primary Sharpe criterion in both universes, and the improvement is statistically significant only on the S&P 500 (ΔSR $= +1.38$, block-bootstrap $p = 0.008$); on the ETF universe the two are indistinguishable ($+0.11$, $p = 0.79$). The remaining methods are not merely unrewarded but actively harmful on the wider universe: PCA ($p = 0.007$), Factor Analysis ($p < 0.001$) and the GKX Autoencoder ($p < 0.001$) are all significantly worse than the uncompressed Baseline on the S&P 500.The paired Rank IC differences are insignificant throughout ($p = 0.74$ on the ETFs, $p = 0.49$ on the S&P 500 for FactorVAE), so what economic advantage exists is not accompanied by a detectable gain in ranking accuracy. The turnover decomposition of Section 4.3 explains why: FactorVAE's edge is largely a persistence effect rather than an accuracy effect, with lower turnover accounting for eight of the nine-and-a-half percentage points by which its annualised alpha exceeds the Baseline's on the S&P 500, and for the whole of its narrower ETF advantage. Gross of transaction costs the compressed representations are approximately flat rather than loss-making, so the honest statement of Q1 is that compression did not add gross predictive value in this setting; where it helped, it did so by producing a ranking that needed to be traded less. The answer to Q1 is therefore conditional rather than general: compression helps only for one representation, on one universe, on one of the two criteria.

Q2: Do deep extractors produce better features than linear ones? Not as a class. The deep/linear axis does not organise the results, because the two deep extractors sit at opposite ends of the ranking: FactorVAE is the least unprofitable configuration in both universes, whereas the GKX Autoencoder is among the weakest and records the worst Sharpe ratio anywhere in the study (-6.510 on the S&P 500 under CatBoost). PCA and Factor Analysis, by contrast, behave stably across folds at a small fraction of the engineering and computational cost.The most likely explanation is regularisation rather than depth: the stochastic latent and KL penalty of the variational model constrain the representation in a way that the deterministic reconstruction objective does not, and in the small-sample cross-sections studied here such a constraint would plausibly be decisive. The turnover evidence gives this conjecture a concrete channel: the regularised representation yields a cross-sectional ordering that is markedly more stable through time ($T \approx 0.9$ against ≈ 1.7 for the deterministic autoencoder), and it is this stability, rather than any gain in point-in-time ranking accuracy, that the net Sharpe ratio rewards. The design does not isolate this mechanism, however—FactorVAE also differs in its recurrent encoder and its use of a look-back window—so the attribution remains a conjecture that a representation-level analysis would be needed to confirm. Architectural complexity therefore does not justify its cost on its own terms; it does so only where it is accompanied by regularisation appropriate to the sample size. This comparison is qualified by the two-track tuning protocol of Section 3.3: the linear reducers were tuned jointly with the downstream model, whereas the deep extractors were selected on a standalone criterion and then frozen, so the deep/linear contrast is not fully controlled.

Q3: How does normalisation interact with method and universe breadth? The choice is not neutral, and its effect differs by criterion and by universe. Cross-sectional normalisation raises the Rank IC in eight of the twelve linear-configuration comparisons, and in all six of those run under CatBoost—where PCA on the ETF universe improves from +0.0031 to +0.0153 and Factor Analysis on the S&P 500 moves from −0.0010 to +0.0015—which is the expected consequence of removing the time-common component from a rank-based task. That gain converts into better portfolio economics only on the wider universe: on the S&P 500 demeaning improves the Sharpe ratio (Factor Analysis under CatBoost from −4.866 to −3.920), while on the sixty-instrument ETF universe it frequently worsens it (the Baseline under LightGBM from −0.648 to −1.333), because the daily cross-sectional moments are estimated from too few names and the resulting estimation error is injected into every feature. The preference inverts once more for the standalone deep signals, where the cross-sectional level of the latent factor is itself informative and standard normalisation dominates on the ETF universe (−0.248 against −0.944 for FactorVAE). Normalisation must therefore be chosen jointly with reference to the breadth of the universe and to whether the signal is consumed as a ranking or as an expected-return level.

5.2 Contributions and Practical Implications

With the research questions answered, the results reported above can be placed in the context of the wider literature and their practical consequences drawn out. The contribution of this thesis is as much methodological as it is empirical. By evaluating PCA, Factor Analysis, the GKX Autoencoder, and FactorVAE as interchangeable upstream modules feeding a fixed downstream predictor, the study isolates the value of the learned representation from the value of the prediction head to which it has previously been welded in the literature. The GKX Autoencoder and FactorVAE were originally proposed and assessed as end-to-end asset pricing models; recasting them as general-purpose feature extractors and comparing them head to head with linear methods on a common pipeline is the principal novelty of the work. The strict walk-forward framework, in which every preprocessing transformation—standardisation, the PCA and FA projections, and the neural network weights alike—is fitted only on training data within each fold and applied without refitting, removes a source of information leakage that compromises many comparable studies and lends credibility to the out-of-sample figures reported.

For a practitioner, the results carry a clear and somewhat sobering message. Dimensionality reduction should be treated as a hypothesis to be tested on the specific universe and signal library at hand, not as a default preprocessing step adopted on theoretical grounds.The evidence does not support a blanket verdict on deep extractors: the deterministic autoencoder is difficult to justify at this data scale, where gradient boosting already absorbs redundant inputs gracefully and where simpler linear compression, or no compression at all, offers comparable behaviour at a fraction of the complexity, whereas the regularised variational model is the one representation whose additional cost is arguably repaid. Where a compression stage is adopted, it should be selected and validated against the economic objective that the strategy ultimately serves, with the reduction method, the downstream model, and the portfolio construction rule evaluated as a single system rather than optimised in isolation.

5.3 Limitations

Several limitations bound the scope of these conclusions and should temper their generalisation. The first concerns data. The feature library is restricted to eighty alpha signals derived solely from price and volume data, and the universes, while deliberately distinct in microstructure and regime, are modest in cross-sectional breadth — sixty instruments in the ETF case and the large-cap S&P 500 constituents in the other.The ETF universe in particular yields the small-sample regime that disadvantages the deep extractors, so the weak showing of the deterministic autoencoder, and the modest margin by which FactorVAE improves on the Baseline, should both be read as conditional on this data scale rather than as a verdict on deep generative methods in general. A universe with thousands of names, or a feature library incorporating fundamental and alternative data, might shift the balance.

The second limitation concerns the architectural and configuration choices made for the deep extractors. The GKX Autoencoder and FactorVAE were tuned within finite search spaces and selected using a standalone Sharpe-based criterion before being passed to the downstream models; a richer search, alternative architectures, or a different selection rule could alter their relative standing. The S&P 500 universe is built from constituents with a complete price and alpha history over the evaluation window, which introduces both survivorship bias and index-membership look-ahead. The reliance on Yahoo Finance daily data, the specific transaction cost and slippage assumptions, the dollar-neutral long–short construction with a fixed selection percentage, and the execution conventions embedded in the backtester all further shape the reported portfolio economics and constrain how directly the figures translate to live trading. The turnover figures reported in Section 4.3, and the cost drag derived from them, are properties of this daily rebalancing rule and this selection fraction rather than of the representations themselves; a lower-turnover construction would shift the net figures without altering

the gross ones. As with any backtested study, the out-of-sample periods, though genuinely held out under the walk-forward protocol, remain finite historical samples, and the conclusions are properly understood as evidence rather than proof.

5.4 Directions for Future Research

The findings open several avenues for further work. The most direct extension is to scale the empirical design to broader universes and richer feature libraries, both to test whether the deep generative extractors recover an advantage once the small-sample constraint is relaxed and to examine how the methods behave when fundamental, macroeconomic, and alternative data are added to the price- and volume-based signals studied here. Establishing the cross-sectional and feature-space scale at which deep representation learning begins to outperform linear compression would be a valuable contribution in its own right.

A second direction is architectural. The deep extractors are currently aligned with the economic objective only through hyperparameter selection: both are trained on their native losses — mean-squared pricing error against realised returns for the GKX Autoencoder and the evidence lower bound for FactorVAE — and the Sharpe criterion of Section 3.3 enters only as a discrete outer search over finished models, never as gradient information. Embedding a differentiable ranking surrogate directly in the training loss would propagate the portfolio objective into the learned representation itself, narrowing the gap between what the encoder learns to represent and what the portfolio strategy needs. Jointly learning the representation and the ranking objective, while preserving the walk-forward discipline that guards against leakage, is a natural next step. A third avenue concerns regime awareness: because no single method dominated across the two universes and market environments, adaptive or ensemble approaches that select or blend reduction methods according to prevailing market conditions deserve investigation, as does an explicit study of how the relative performance of the extractors evolves through crisis and calm periods.A fourth avenue follows directly from the turnover decomposition. Having established that the realised losses are largely a cost phenomenon — the implied gross alphas are approximately flat — the natural next step is to attack the turnover itself rather than the signal. Varying the selection fraction, imposing a rebalancing band or a no-trade region, smoothing the predicted ranks across days, or penalising turnover directly in the training objective would each test whether an approximately zero gross edge can be converted into a positive net one. That FactorVAE's advantage here is almost entirely a persistence advantage suggests that representations selected explicitly for the temporal stability of the ranking they induce, rather than for point-in-time accuracy, are a promising direction in their own right . Finally, extending the downstream layer beyond tree-based gradient boosting to other model classes would clarify how far the conclusions reached here are properties of the learned features themselves and how far they reflect the particular predictor to which those features were attached.

5.5 Conclusion

Across two asset universes, two downstream learners and two normalisation schemes, no dimensionality reduction method improved on the raw alpha library by a statistically significant margin in both markets. FactorVAE was the sole partial exception: the least unprofitable configuration in each universe, though significantly better than the Baseline only on the S&P 500. Taken together, the evidence in this thesis supports a measured view of dimensionality reduction as alpha. Compression is a tool whose value is real but conditional, not a guaranteed improvement conferred by theory, and the sophistication of the extractor is no substitute for matching the method to the data, the downstream model, and the economic objective it is meant to serve.

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